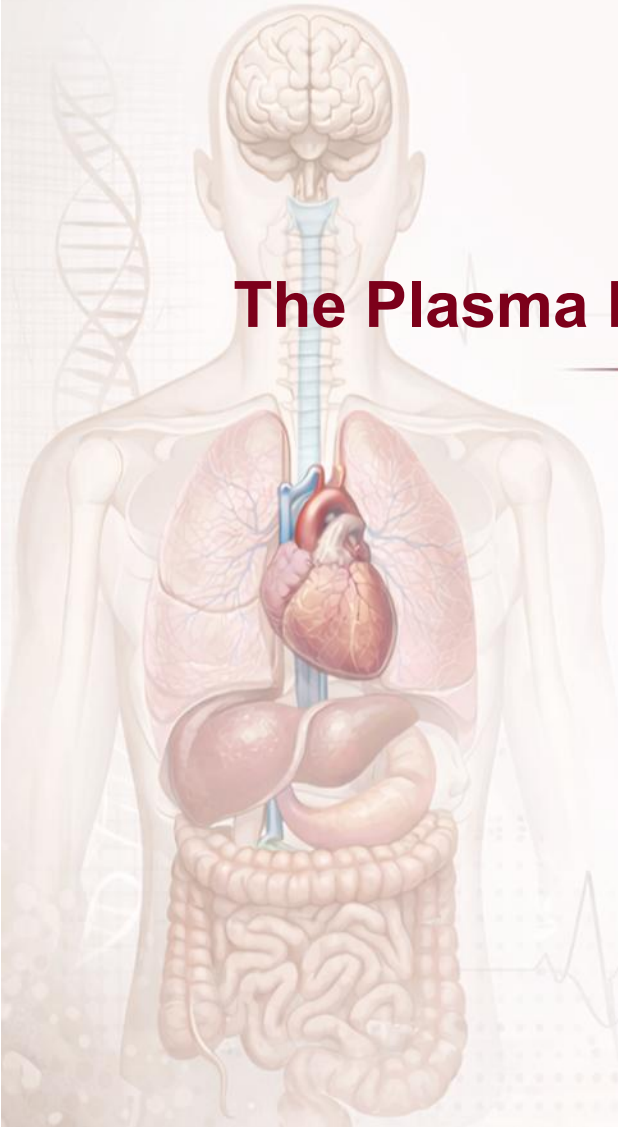


Chapter 3.
The Plasma Membrane and Membrane Potential

KOREA UNIVERSITY
1905

Membrane in terms of homeostasis



https://www.youtube.com/watch?v=tIzF2tWy6KI

Membranes maintain homeostasis

- All cells are enveloped by a plasma membrane, a thin, flexible, lipid barrier separating the cellular contents from its surrounding environment
- Cells exchange materials across this membrane with extracellular fluid
- Plasma membrane contains specific proteins, some of which enables selective passage of materials and *maintain homeostasis*
- Other proteins are receptors for interaction with chemical/protein messengers in the cell to *control cell activity and homeostasis*

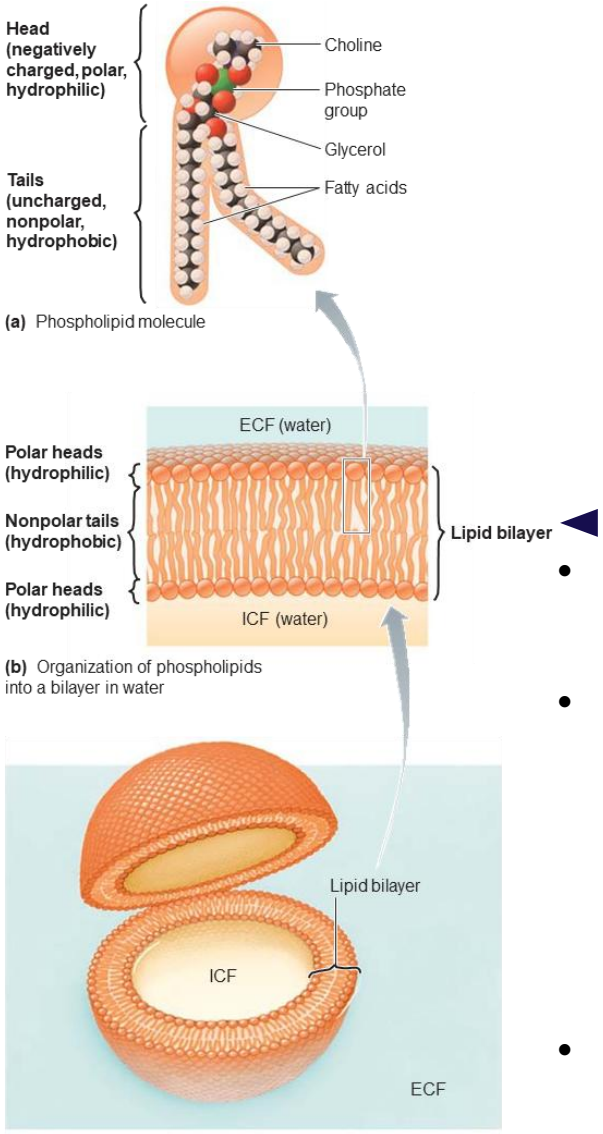
- Cells have membrane potential, which is excess of negative charges lined up inside and excess of positive charges outside the membrane
- Specialized nerve and muscle cells alter their membrane potential on appropriate stimulation, to *maintain whole body homeostasis*

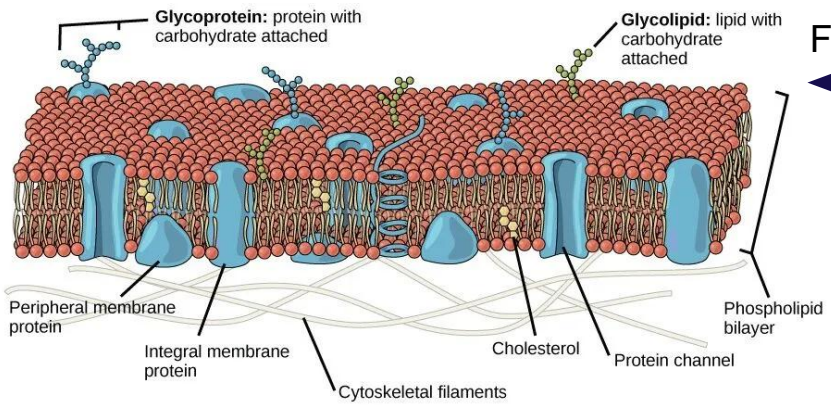
3.1 Membrane Structure and Functions

Membrane structure and functions

- Plasma membrane: extremely thin layer that forms the outer boundary of every cell
  - Encloses the intracellular contents
  - Acts as a mechanical barrier trapping necessary molecules
  - Determines cell composition by selectively permitting specific substances to pass
  - Controls entry of nutrients and molecules and exit of waste products
  - Maintains differences in ion concentrations inside and outside cell
- Plasma membrane is a fluid lipid bilayer embedded with proteins (*fluid mosaic model*)
  - Consists of lipids, proteins, plus small amounts of carbohydrate
  - Phospholipids are the most abundant membrane lipids

Structure and organization of the phospholipids

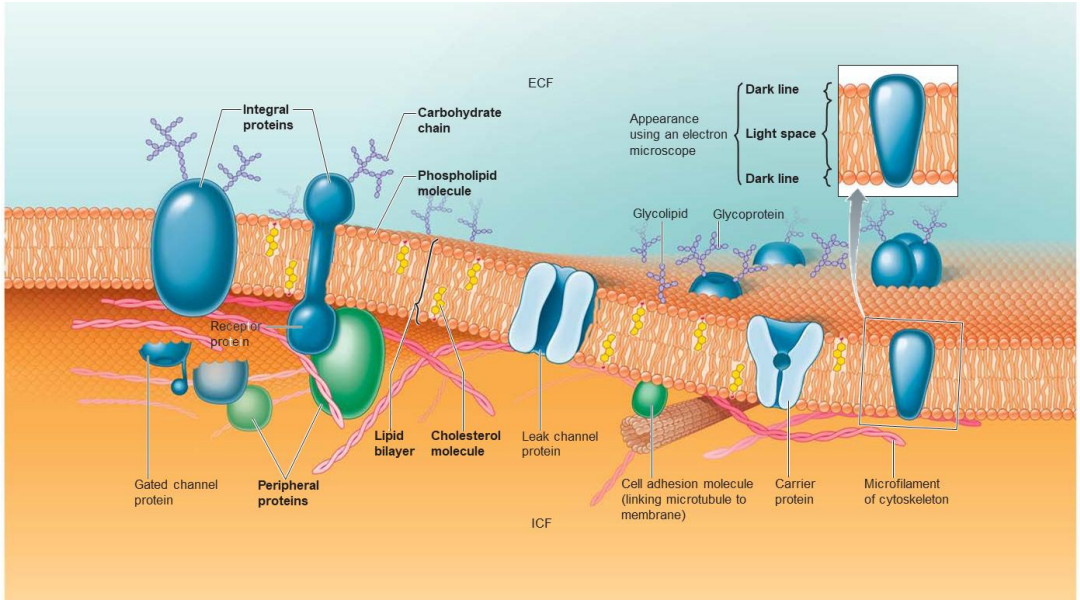




Fluid mosaic model

- (a) Phospholipid example - phosphatidylcholine
- (b) In water, phospholipid molecules self-organize into a lipid bilayer with the polar heads interacting with the polar water molecules at each surface and the nonpolar tails all facing the interior of the bilayer
- (c) An exaggerated view of the plasma membrane enclosing a cell, separating the ICF from the ECF

Other major components of PM than phospholipids



- Cholesterol: fluidity and the stability of the membrane
  - Cholesterols are inserted between the phospholipid molecules
  - Prevent fatty acid chains from packing together and crystallizing
- Membrane proteins are inserted within or attached to the lipid bilayer
  - Integral proteins: monotopic (embedded in only one leaflet of the bilayer) or polytopic (spanning the membrane)
  - Transmembrane proteins: traverse the entire bilayer
  - Peripheral proteins: adhere to the surface, attached to phospholipid heads or integral proteins through non-covalent interactions

Functions of plasma membrane

- Lipid bilayer
  - Forms basic membrane structure
  - Responsible for membrane fluidity (<-> rigidity)
  - Hydrophobic inside blocks the flow of water-soluble substances between ICF and ECF

- Membrane proteins with specialized functions
  - Channels for passage of ions
  - Carriers for transport of substances
  - Docking-marker acceptors for secretory vesicles
  - Receptors responsible for chemical/protein messengers
  - Cell adhesion molecules holding cells together

- Membrane carbohydrates
  - self-recognition markers, cell-to-cell interactions, involved in tissue growth

3.2 Cell-to-Cell Adhesions

Cell-to-cell adhesions

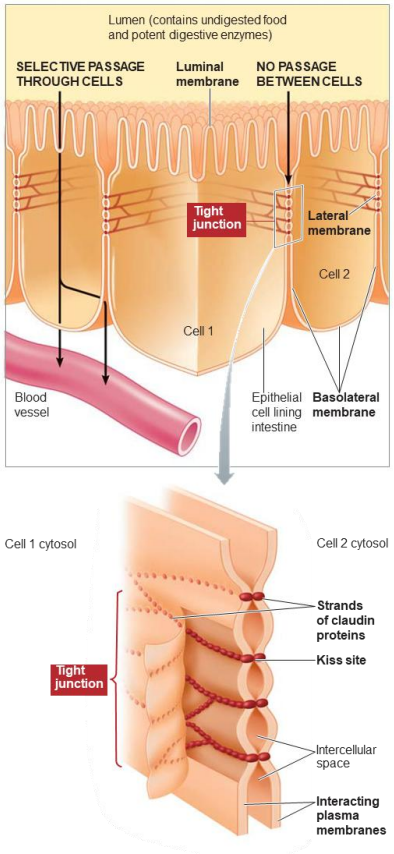
- Extracellular matrix binds cells together
  - Consists of collagen, elastin, and fibronectin
- Some cells are directly linked by specialized cell junctions
  - Desmosomes, tight junctions, and gap junctions

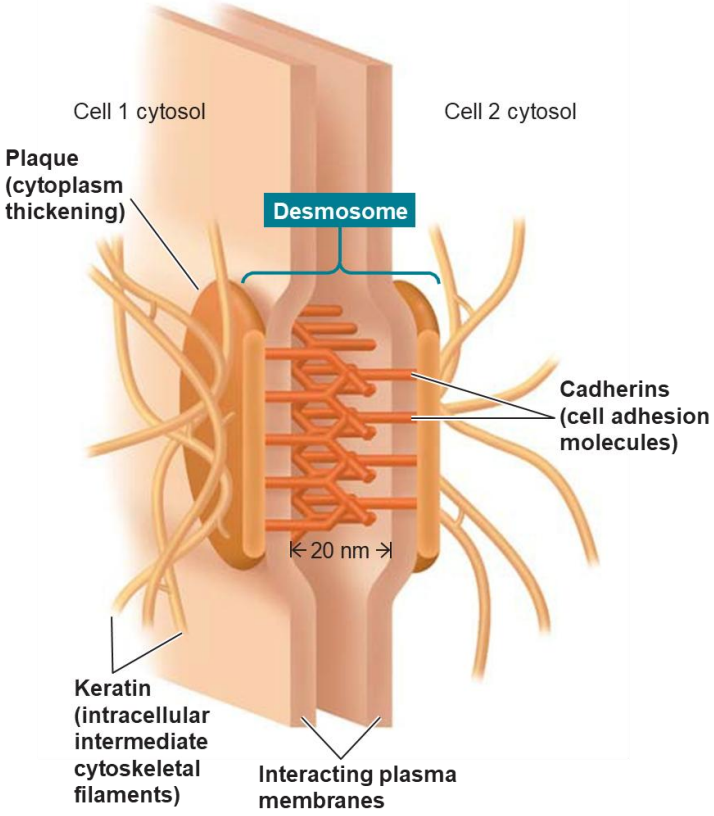
Cell junctions

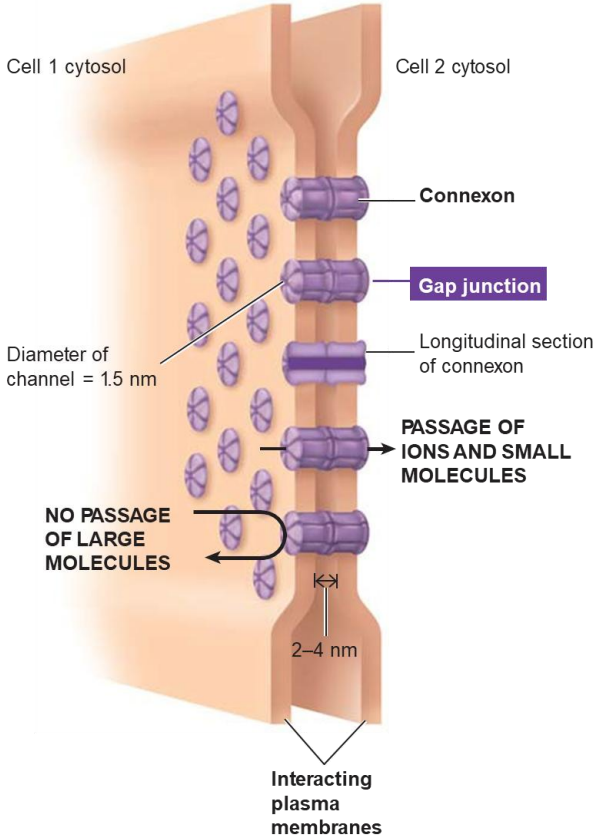


https://www.youtube.com/watch?v=pVWQm-GYK_Y

Cell junctions







1. Tight junction
- impermeable, sealing, prevent the movement of materials
- epithelial tissues
- separates two compartment with different composition
- bladder, intestines, kidney

2. Desmosome
- Anchors two cells together, water/ion-permeable
- tissues under stress -> stretching
- skin, intestines

3. Gap junctions
- Intercellular “tunnels” for water or ions
- cells spreading action potential
- cardiac muscle, neurons
- Communicating junctions, permit movement

3.3 Overview of Membrane Transport

Overview of membrane transport

- There are
  - Substances that can cross the membrane (membrane is permeable): lipid-soluble substances, small water-soluble substances
  - Substances that cannot cross the membrane (membrane is impermeable): low lipid solubility, large -> require assisted transport
  - Substances that can selectively cross the membrane (membrane allow some particles to pass while preventing the others): specific ions or molecules through channels or carriers

During membrane transport

- Lipid-soluble substances and small water-soluble substances can permeate the plasma membrane without assistance
  - Lipid-soluble molecules can pass through lipid bilayer
  - Uncharged or nonpolar molecules like O_2, CO_2, fatty acids

In other cases,

  - For water-soluble (lipid-insoluble) particles like ions (Na^+, K^+) -> transport through protein channels
  - For large and lipid-insoluble particles like glucose -> carrier-mediated assisted transport (facilitated diffusion)

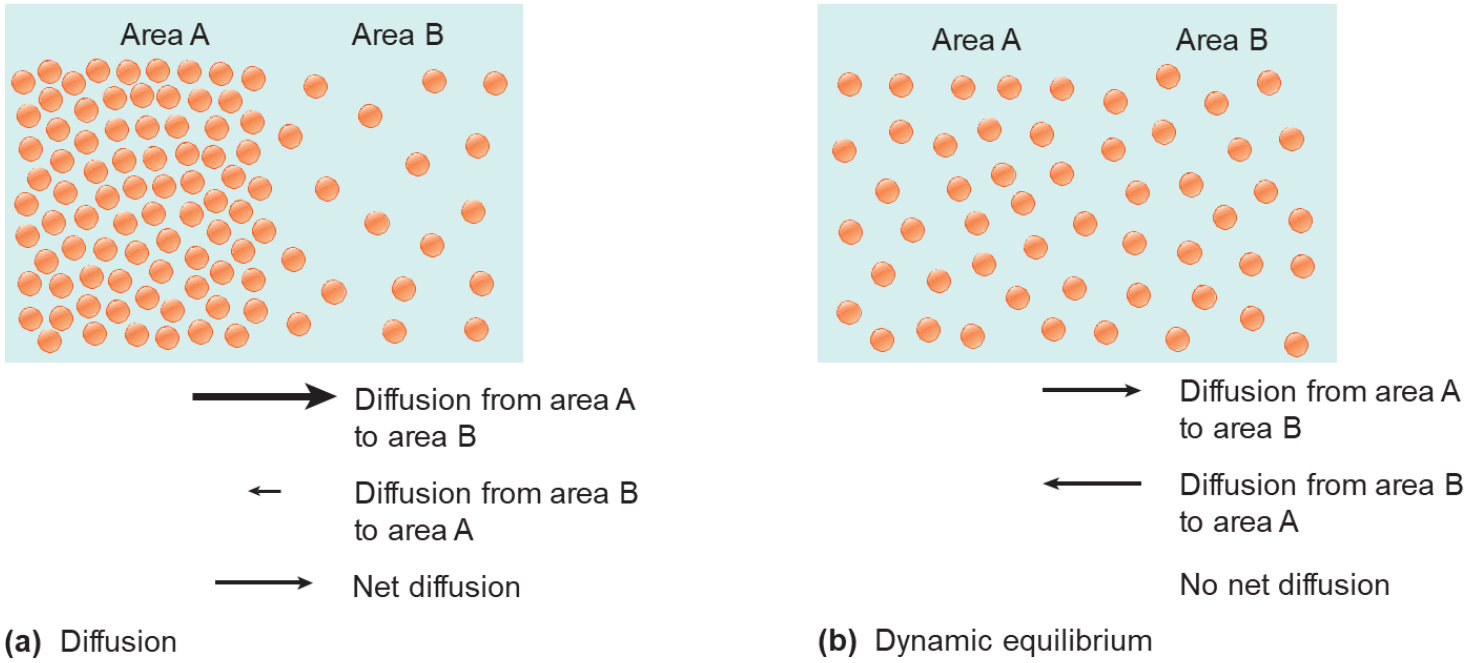
- Some kind of active force is needed to move them across the membrane
  - Passive forces: do not require the cell to expend energy to produce movement
  - Active forces: do require the cell to expend energy (ATP) in transporting a substance across the membrane

Summary

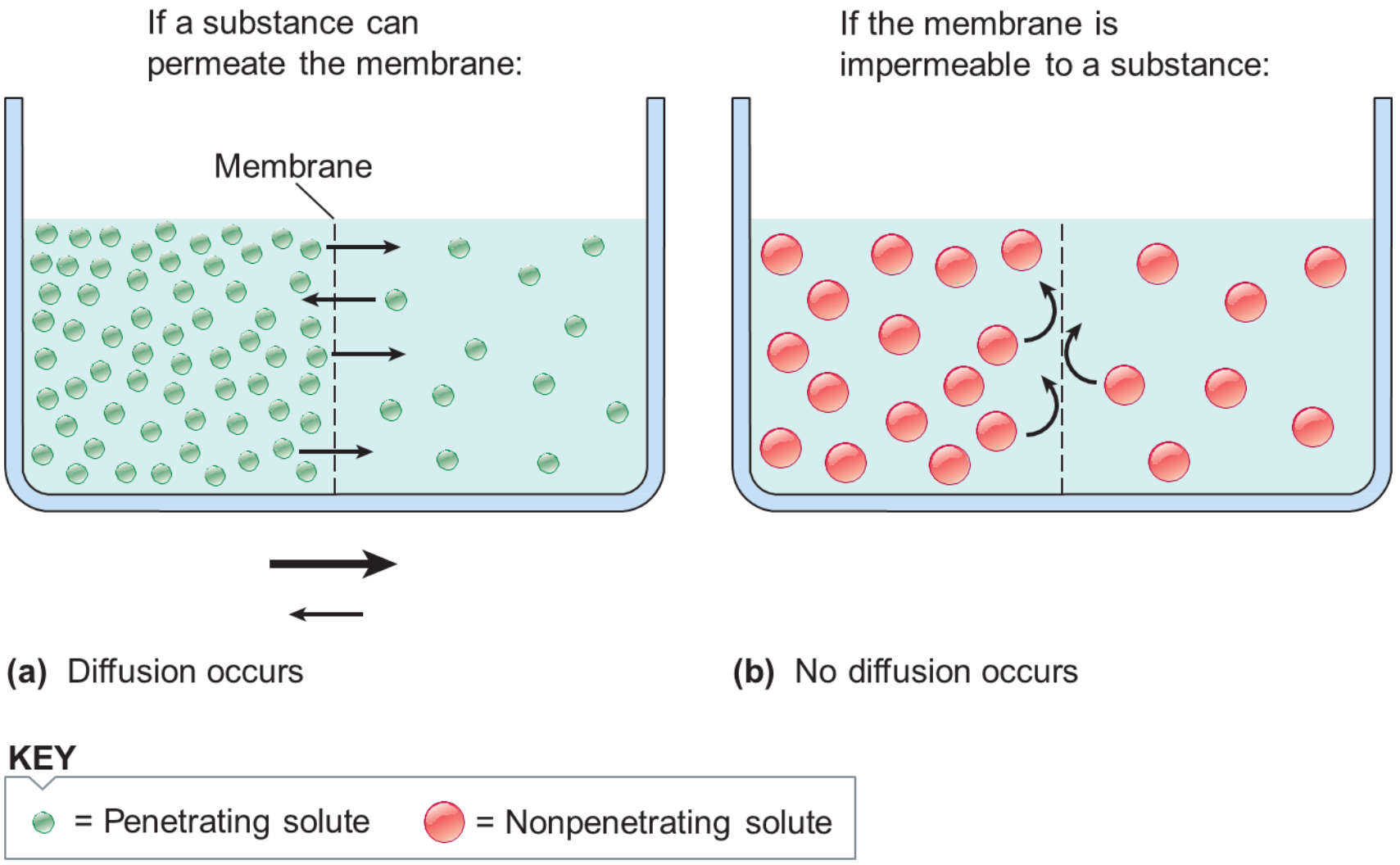
- The plasma membrane is composed of lipid bilayer
- Due to the structure of the membrane, it is selectively permeable
- Molecules which are not permeable to the membrane can be transported using assisting molecules

3.4 Unassisted Membrane Transport

Unassisted membrane transport



Unassisted membrane transport



Unassisted membrane transport – chemical gradient

- Particles that can penetrate the membrane on their own are passively driven across the membrane
  - Diffusion down a concentration gradient: molecules tend to become evenly distributed over time
  - Simple diffusion: uniform spreading out of molecules as a result of random intermingling
  - *Chemical gradient*: difference in concentration of chemical substance between two adjacent areas
  - Net diffusion: difference between two opposing diffusion (10-2=8)
  - *Fick's Law of Diffusion*: rate of net diffusion is affected by factors including
    - Concentration gradient of substance
    - Surface area of membrane
    - Lipid solubility



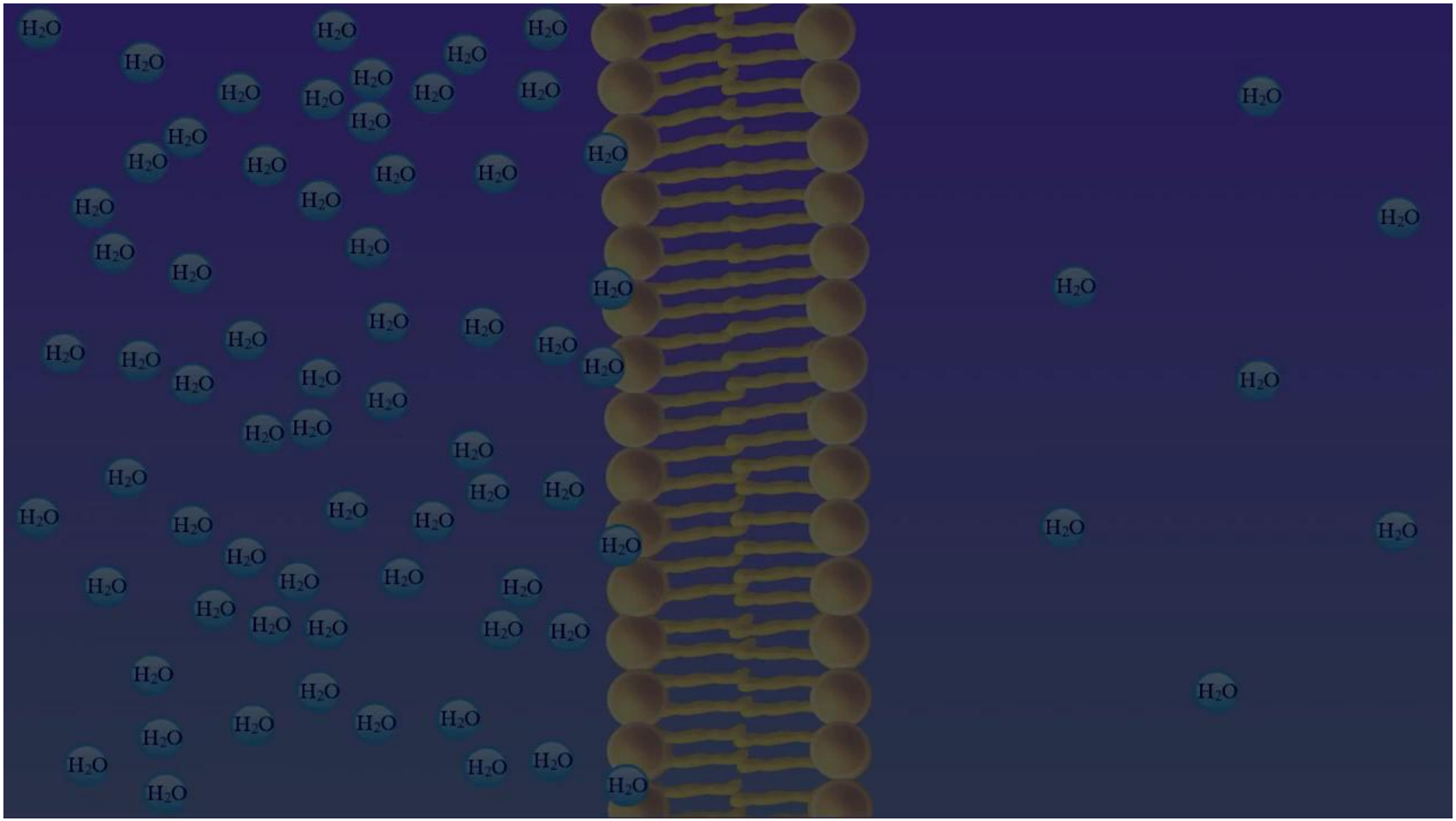
- Molecular weight of substance
- distance



Unassisted membrane transport – electrical gradient

- Ions also can penetrate the membrane passively along their electrical gradient
  - *Electrical gradient* is created by positive and negative charge on ions
  - Ions with like charges tend to repel each other
  - Positively charged ions (cations) tend to move toward the more negatively charged area
  - A difference in charge between two adjacent area creates an electrical gradient promoting ion movement to the opposite charge area
- Net of chemical and electrical gradient = electrochemical gradient

Unassisted membrane transport – osmosis

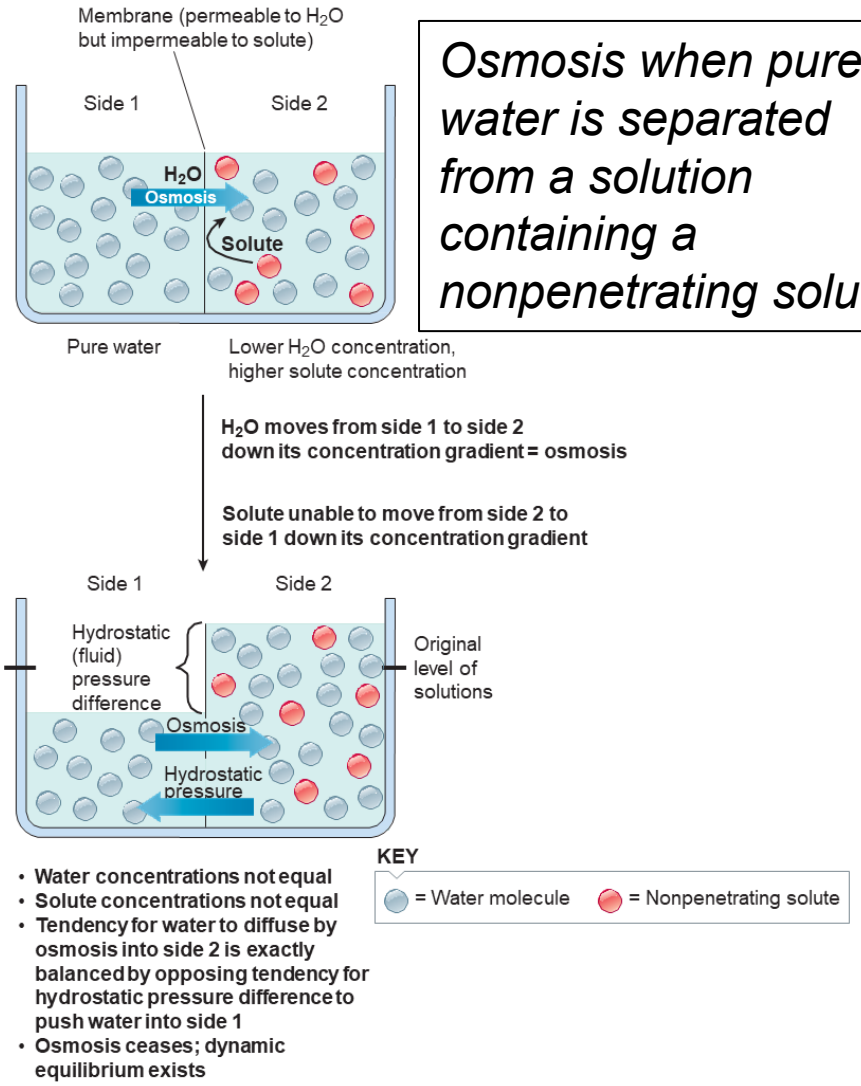


https://www.youtube.com/watch?v=BDDqY7IZhxQ

Unassisted membrane transport – osmosis

- Osmosis: the net diffusion of <u>water</u> down its concentration gradient

- Tonicity: a measure of the effective osmotic pressure gradient; the water potential of two solutions separated by a partially-permeable cell membrane
  - Determines whether cell remains same size, swells, or shrinks when a solution surrounds the cell

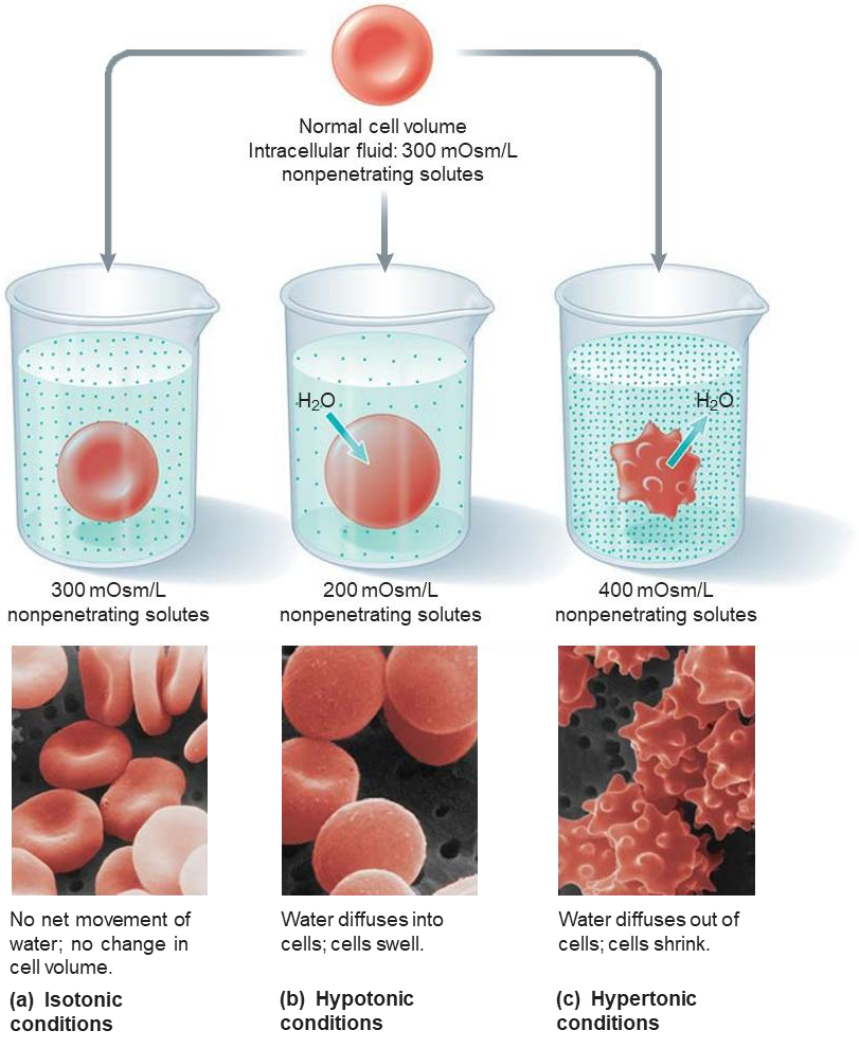


Osmosis when pure water is separated from a solution containing a nonpenetrating solute

Unassisted membrane transport – osmosis

- Tonicity: a measure of the effective osmotic pressure gradient; the water potential of two solutions separated by a partially-permeable cell membrane
  - Isotonic solution: same concentration of nonpenetrating solutes as normal body cells
  - Hypotonic solution: below-normal concentration of nonpenetrating solutes
  - Hypertonic solution: above-normal concentration of nonpenetrating solutes

Unassisted membrane transport – osmosis

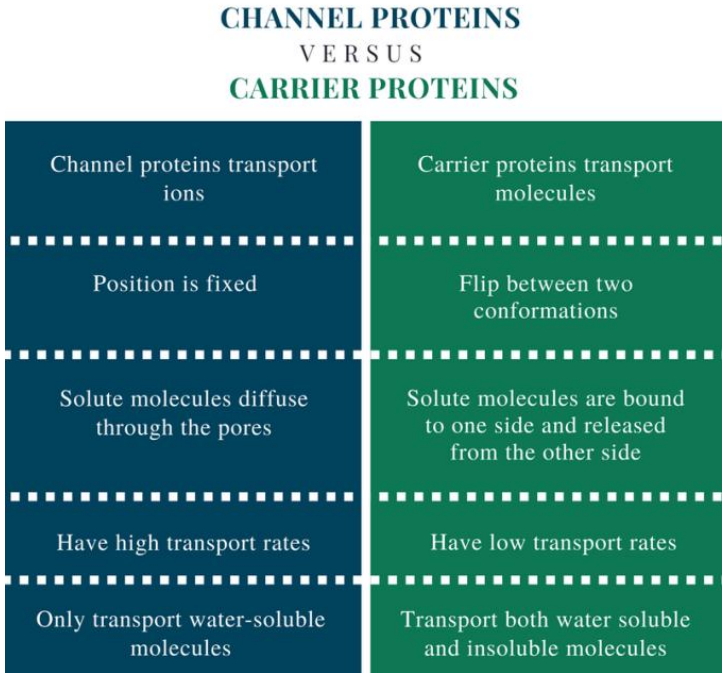


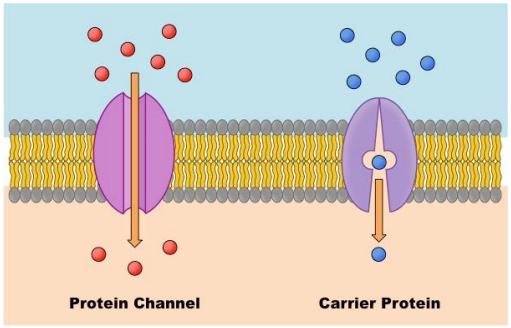
3.5 Assisted Membrane Transport

1. Carrier-mediated transport
2. Passive diffusion (carrier-mediated)
3. Active transport (carrier-mediated)
4. Vesicular transport

Channel vs Carrier

- Major transport proteins
  - Channel and carrier proteins are both integral membrane proteins that facilitate substance movement across cell membranes
  - Channels form passive, open pores for rapid ion movement
  - Carriers (=Transporters, 막 수송체) bind specific molecules, change shape, and can perform slower, active, or passive transport, often against concentration gradients

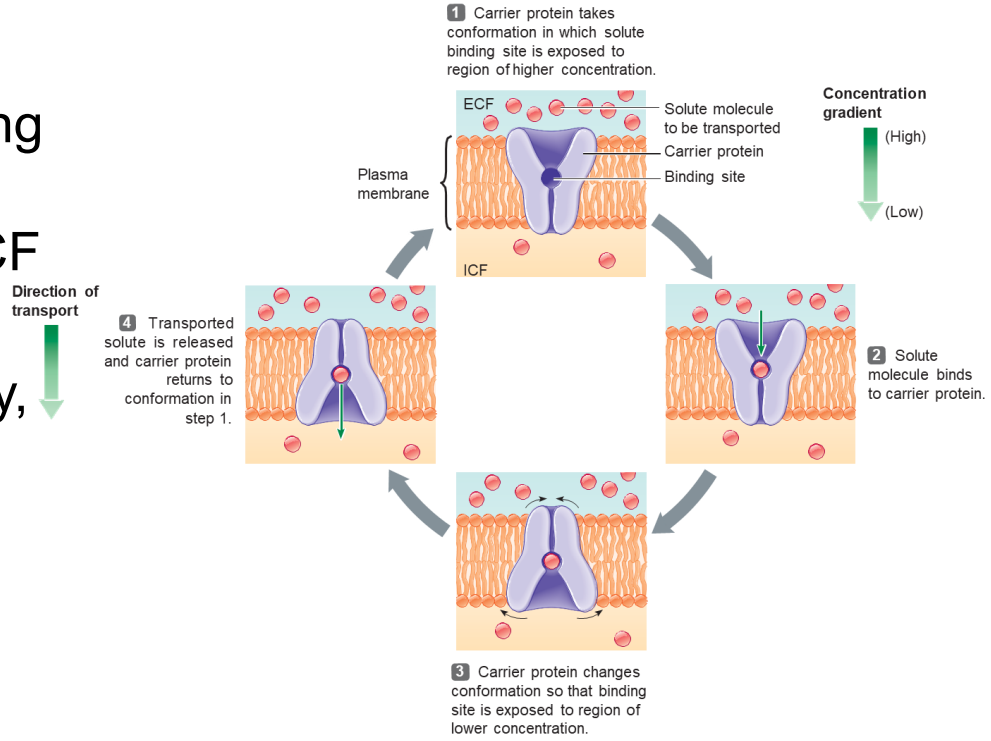




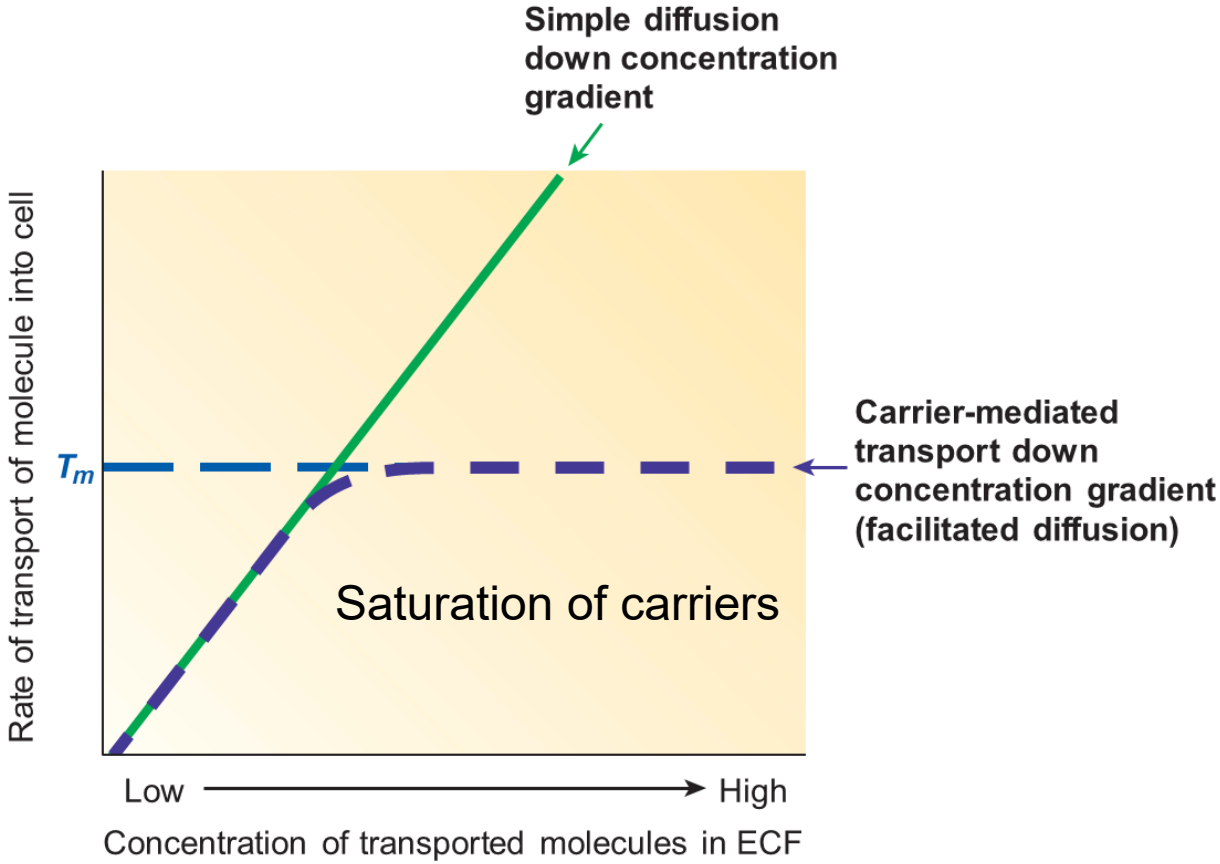
https://pediaa.com/difference-between-channel-and-carrier-proteins/

Assisted membrane transport

- Carrier-mediated transport
  - Carrier protein spans the thickness of the plasma membrane
  - Can change its conformation (shape) so that specific binding sites within the carrier are alternately exposed to the ECF and the ICF
  - Key characteristics: specificity, saturation, and competition



Simple diffusion vs Carrier-mediated transport



Simple diffusion of a molecule down its concentration gradient: the rate of transport into the cell is directly proportional to its extracellular concentration.
Carrier-mediated transport: the rate of transport into the cell is directly proportional to the extracellular concentration of the molecule until the carrier is saturated, at which time the rate of transport reaches the transport maximum (Tm). After Tm is reached, the rate of transport levels off despite further increases in the ECF concentration of the molecule.

Assisted membrane transport

- Facilitated diffusion is passive carrier-mediated transport
  - Uses a carrier to facilitate the transfer of a particular substance across the membrane “downhill” from high to low concentration (no ATP)
- Active transport is carrier-mediated and uses energy
  - *Primary active transport* directly uses ATP to power transport (Na^+-K^+ ATPase pump)
  - *Secondary active transport* uses the electrochemical gradient created by primary transport as its energy source to move substances such as glucose, via symporters or antiporters (sodium and glucose transporter, SGLT)
- Vesicular transport: material is moved in and out of the cell wrapped in membrane (endocytosis and exocytosis)
  - Rates of endocytosis and exocytosis must be kept in balance to maintain a constant membrane surface area

Active transport – primary and secondary

Active Cellular Transport:
Primary & Secondary

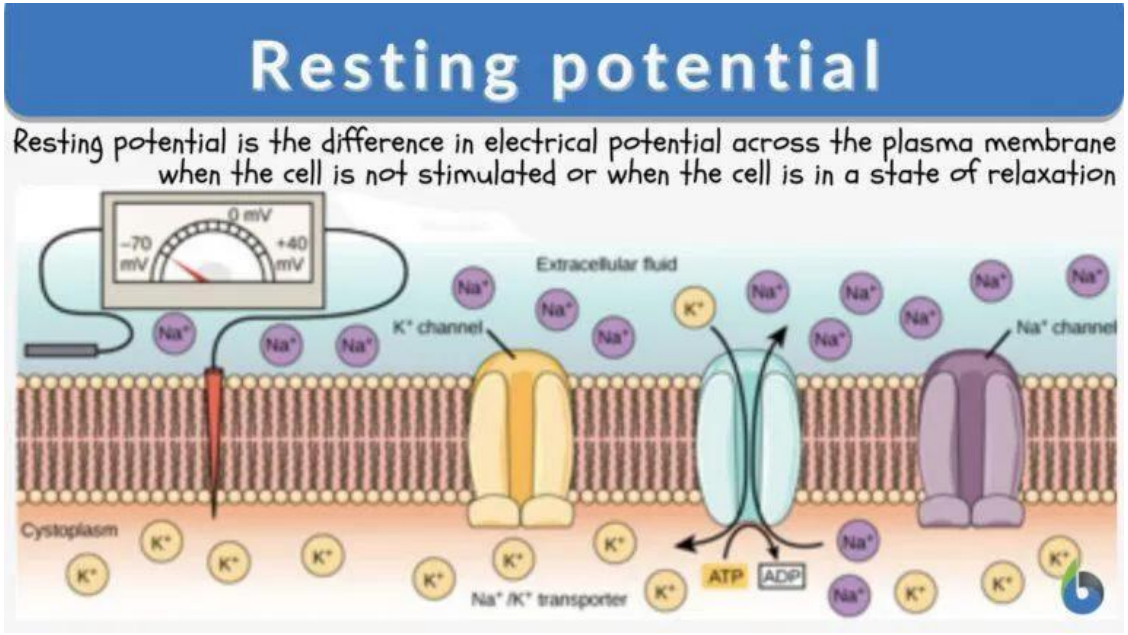
Primary and secondary transporters
Uniport, symport, and antiport

https://www.youtube.com/watch?v=BDDqY7IZhxQ

3.6 Membrane Potential

Membrane potential

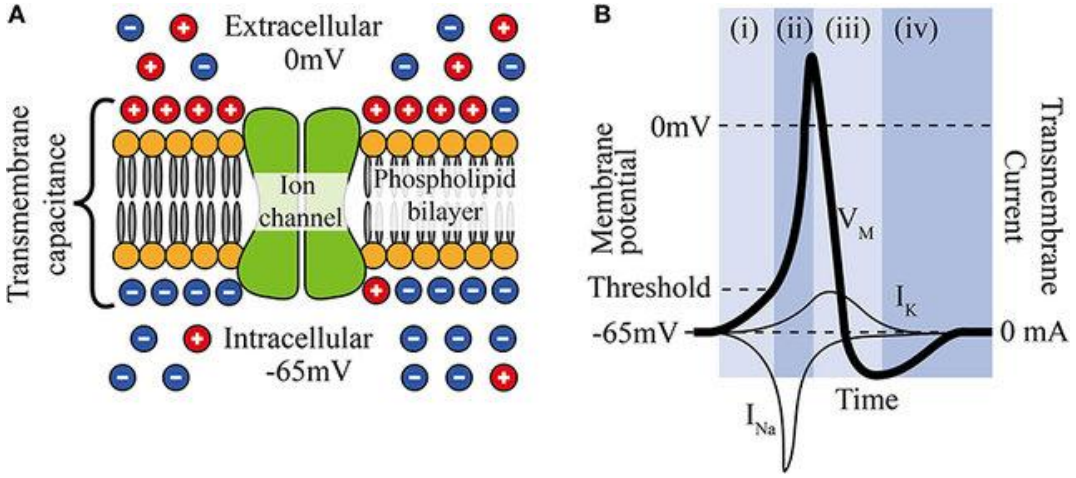
- Cell plasma membranes are electrically polarized
  - In other words, membrane potential is a separation of opposite charges across the plasma membrane (between ICF and ECF)
  - This is created by differences in the concentration and permeability of key ions (K^+ > Na^+ > …)



https://www.biologyonline.com/dictionary/resting-potential

Membrane potential

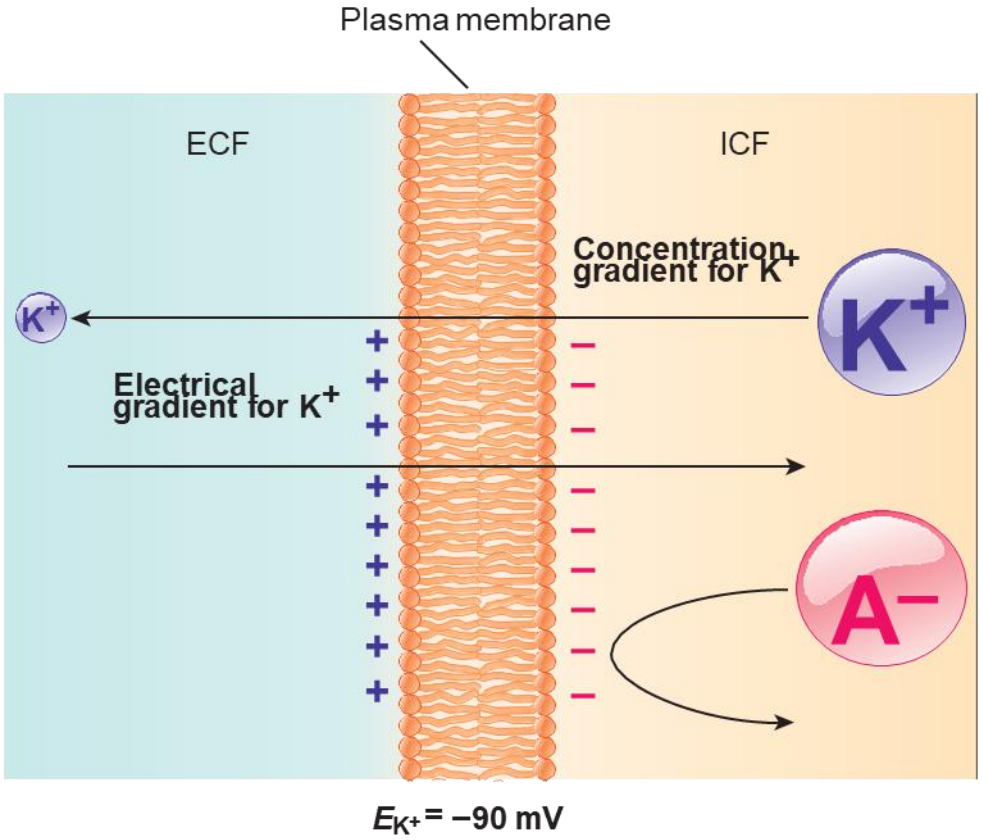
- Cell plasma membranes are electrically polarized
  - Although all cells have membrane potential (mV unit), the cells of excitable tissues (neuron, muscle cells) can produce transient change in their membrane potential (*action potential*)
  - Action potentials are responsible for producing nerve impulses in nerve cells and for triggering contraction in muscle



https://www.frontiersin.org/journals/neuroengineering/articles/10.3389/fneng.2014.00027/full

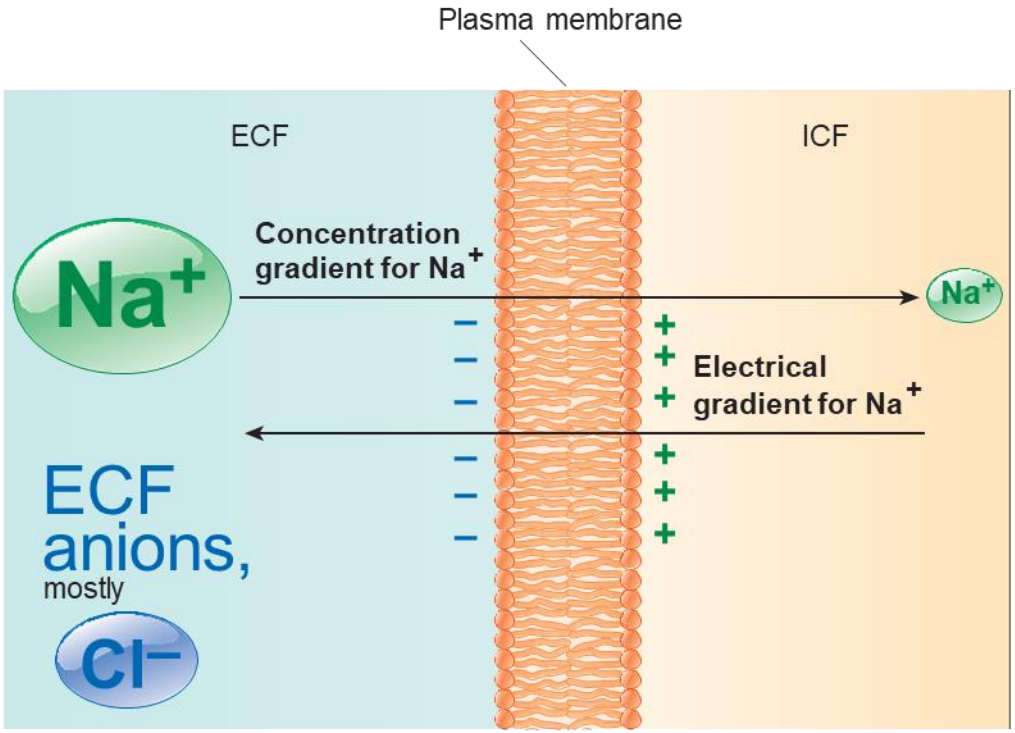
Equilibrium potential for K^+

Equilibrium potential (E_{ion}) is the membrane potential at which the electrical gradient exactly balances the concentration gradient for a specific ion, resulting in no net movement of that ion across a membrane.



1. The concentration gradient for K^+ tends to move this ion out of the cell.
2. The outside of the cell becomes more positive as K^+ moves to the outside down its concentration gradient.
3. The membrane is impermeable to the large intracellular protein anion (A^-). The inside of the cell becomes more negative as K^+ moves out, leaving behind A^-.
4. The resulting electrical gradient tends to move K^+ into the cell.
5. No further net movement of K^+ occurs when the inward electrical gradient exactly counterbalances the outward concentration gradient. The membrane potential at this equilibrium point is the equilibrium potential for K^+ (E_{K^+}) at –90 mV.

Equilibrium potential for Na^+



1. The concentration gradient for Na^+ tends to move this ion into the cell.
2. The inside of the cell becomes more positive as Na^+ moves to the inside down its concentration gradient.
3. The outside becomes more negative as Na^+ moves in, leaving behind in the ECF unbalanced negatively charged ions, mostly Cl^-.
4. The resulting electrical gradient tends to move Na^+ out of the cell.
5. No further net movement of Na^+ occurs when the outward electrical gradient exactly counterbalances the inward concentration gradient. The membrane potential at this equilibrium point is the equilibrium potential for Na^+ (E_{Na^+}) at +60 mV.

E_{Na^+} = +60 mV

The Nernst equation calculates the equilibrium potential of a specific ion (Eion) based on its concentration difference across the membrane.

Equation (at 37°C): Eion = (61 / z) log([ion]out / [ion]in)
Eion: equilibrium potential (mV)
z: valence (charge) of the ion (Na+ = +1, K+ = +1, Cl- = -1)
[ion]out: extracellular ion concentration, [ion]in: intracellular ion concentration

The equilibrium potential values for each ion (Eion) are determined by the Nernst equation and reflect the opposing forces acting on each ion.

Effect of concurrent K^+ and Na^+ movement on establishing the resting potential

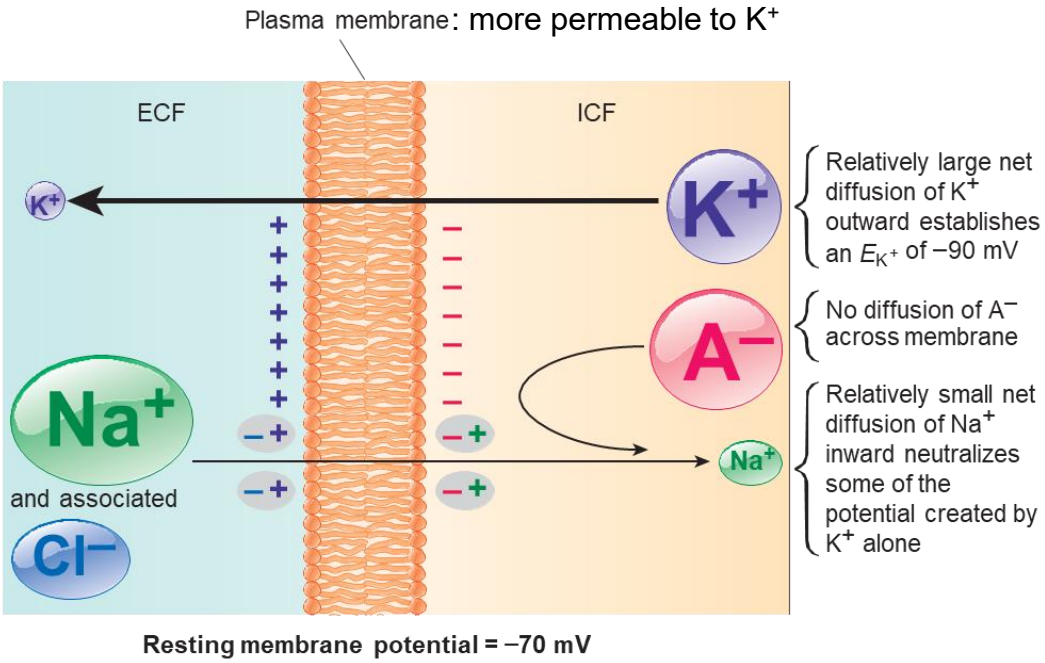
1 The Na^+–K^+ pump actively transports Na^+ out of and K^+ into the cell, keeping the concentration of Na^+ high in the ECF and the concentration of K^+ high in the ICF.

2 Given the concentration gradients that exist across the plasma membrane, K^+ tends to drive membrane potential to the equilibrium potential for K^+ (–90 mV), whereas Na^+ tends to drive membrane potential to the equilibrium potential for Na^+ (+60 mV).

3 However, K^+ exerts the dominant effect on resting membrane potential because the membrane is more permeable to K^+. As a result, resting potential (–70 mV) is much closer to E_{K^+} than to E_{Na^+}.

4 During the establishment of resting potential, the relatively large net diffusion of K^+ outward does not produce a potential of –90 mV because the resting membrane is slightly permeable to Na^+ and the relatively small net diffusion of Na^+ inward neutralizes (in *gray* shading) some of the potential that would be created by K^+ alone, bringing resting potential to –70 mV, slightly less than E_{K^+}.

5 The negatively charged intracellular proteins (A^-) that cannot cross the membrane remain unbalanced inside the cell during the net outward movement of the positively charged ions, so the inside of the cell is more negative than the outside.



Resting membrane potential = –70 mV

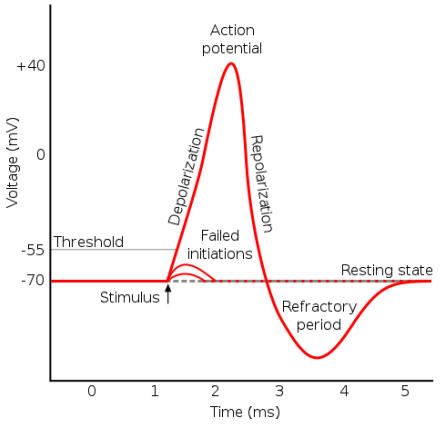
Membrane potential



https://www.youtube.com/watch?v=MplWXZTOk6o

Resting membrane potential

- The resting membrane potential (RMP)
  - The stable electrical potential difference across the plasma membrane of cell when it is not actively generating an action potential, typically around −70 mV in neurons
  - It arises from the unequal distribution of ions across the membrane and the selective permeability of the membrane, which is much higher for K^+ than for Na^+ due to abundant K^+ leak channels
  - As a result, K^+ efflux plays a dominant role in making the inside of the cell negative, while a small Na^+ influx slightly offsets this effect



- The resting membrane potential is close to EK^+ value because the cell membrane is much more permeable to K^+ than to Na^+, due to the abundance of K^+ leak channels